SILICON PHOTONICS
FOR THE SYSTEM ARCHITECT

A Role A Tutorial: Physical Layer to Platform Architecture

Targeting x86/BIOS Engineers Transitioning to AI Datacenter Platforms

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For Technical Architects in AI Infrastructure

Preface: Why a System Architect Needs This

Silicon photonics is no longer a research topic. As of 2026, it is shipping in production datacenter hardware. NVIDIA's Quantum-X and Spectrum-X switches integrate co-packaged optics directly with switch ASICs. TSMC's COUPE platform is in active deployment. Broadcom's Tomahawk 5-Bailly is the industry's first volume-production co-packaged optics solution. Large-scale adoption is projected between 2028 and 2030.

This creates a hard requirement: any architect defining AI datacenter platforms — compute nodes, fabric topology, SoC I/O boundaries, or thermal/power envelopes — must understand silicon photonics at a system level. Not to design photonic devices, but to make informed decisions about packaging constraints, I/O bandwidth allocation, thermal co-design, and interface boundaries.

This tutorial is written for engineers with deep x86/SoC/firmware backgrounds who have no photonics training. The goal is working fluency at the system architecture level in 8–12 weeks of part-time study.

> **ARCHITECT NOTE:** This tutorial covers Role A: System/Platform Architect. It does not target Role B (photonic device or PIC design), which requires a foundation in electromagnetics and waveguide physics and a multi-year ramp.

> **ARM NEOVERSE:** Throughout this document, ARM Neoverse-specific callouts show how each concept applies to Neoverse N2/V2/V3 and CMN-based datacenter platforms — the primary target for next-generation optical interconnect integration.

Chapter 1: Why Photonics — The Electrical Wall

1.1 The Copper Bottleneck

Electrical signaling over copper traces is governed by fundamental physics. As data rates increase, three effects dominate:

- Skin effect: at high frequencies, current crowds to the conductor surface, increasing effective resistance.
- Dielectric loss: signal energy is absorbed by the PCB substrate, worsening with frequency.
- Impedance matching and crosstalk: at 112G and 224G SerDes speeds, maintaining signal integrity across board-level distances becomes increasingly expensive in power and silicon area.

The consequence: a traditional pluggable transceiver in a 400G or 800G module requires a digital signal processor (DSP) chip to clean up signal degradation. Each DSP adds latency, consumes 10–20W, and introduces an additional conversion step.

> **SCALE IMPACT:** In a 1-million-GPU cluster at 6 transceivers per GPU, DSP-based pluggable optics would consume approximately 180MW from transceiver overhead alone.

1.2 What Light Changes

Optical signals propagate with fundamentally different characteristics:

Property	Copper (224G SerDes)	Silicon Photonics
Loss mechanism	Resistance + dielectric	Waveguide scattering (low)
Bandwidth scaling	Degrades with distance	Near-constant with distance
DSP requirement	Required at 224G+	Eliminated with CPO
Power/bit	~5–8 pJ/bit (board-level)	~1–2 pJ/bit (target)
Reach	<1m at 224G	cm to km

1.3 The Architecture Consequence

From a system architect's view, the transition to optical interconnect is not just a physical layer change. It restructures where bandwidth exists, how power is allocated, and what the packaging boundary looks like. The traditional model — ASIC + pluggable transceiver + fiber — is being replaced by a model where the optical engine is co-located with the ASIC die, inside the same package.

This changes thermal maps, mechanical constraints, power delivery domains, and firmware surface area. Understanding these changes is the core of Role A competency.

> **ARM NEOVERSE:** Neoverse V3 and CMN-700/CMN-700AE mesh interconnects currently assume electrical SerDes for die-to-die and board-level interfaces. As CPO matures, the question for Neoverse platform architects is: where does the optical boundary enter the package, and what does firmware need to know about it?

Chapter 2: Silicon Photonics Fundamentals — What the Architect Must Know

2.1 The Core Principle

Silicon photonics uses near-infrared light (typically 1310nm or 1550nm wavelength) to carry data. The silicon chip acts as a waveguide medium — light travels through microscopic channels etched into the silicon substrate, guided by the difference in refractive index between silicon (n ≈ 3.5) and the surrounding silicon dioxide cladding (n ≈ 1.5).

The chain from electrical to optical and back is:

Laser Source → Modulator → Waveguide → Photodetector → Electrical Output

As a system architect, you do not need to design these components. You need to understand their constraints, failure modes, and system implications.

2.2 Key Devices and Their Architectural Implications

Waveguides

Silicon waveguides confine and route light on-chip. Typical cross-sections are ~450nm wide — sub-micron structures fabricated with standard CMOS processes. Waveguide loss is measured in dB/cm. For an architect, the key implication is that routing optical paths on a photonic integrated circuit (PIC) is a layout constraint, not unlike metal routing in an electronic IC, but with tighter bend radius rules.

> **ARCHITECT NOTE:** Waveguide loss budgeting is the optical analog of IR drop analysis. A silicon photonics architect must track optical loss across all path elements — couplers, splitters, bends, connectors — just as a power architect tracks resistive loss in a PDN.

Modulators

The modulator encodes data onto light by varying its intensity or phase. Two dominant types in silicon photonics:

- Mach-Zehnder Modulator (MZM): Splits light into two paths, shifts phase of one, then recombines. Large footprint (~1mm), broad bandwidth, temperature-stable. Used in coherent transceivers and high-baud-rate links.
- Ring Resonator Modulator: Compact (~10µm diameter), low power, but highly temperature-sensitive. A 1°C temperature shift can detune a ring resonator. Used in CPO optical engines at shorter reaches.

> **ARCHITECT NOTE:** Ring resonator thermal sensitivity is a first-order concern for architects defining thermal envelopes. A CPO device with ring modulators has hard thermal constraints that must be co-planned with the package thermal budget. This is not a device-level problem — it is a platform-level problem.

Photodetectors

Photodetectors convert optical signals back to electrical. Silicon photonics platforms use germanium-on-silicon (Ge-on-Si) photodetectors. Germanium absorbs near-infrared light efficiently. Key specifications for architects: responsivity (A/W), bandwidth (GHz), and dark

current. From a system perspective, the photodetector defines the receiver sensitivity floor and contributes to the link power budget.

Optical Couplers

Light must enter and exit the photonic chip. Couplers are the interface between the PIC and the external fiber or fiber array. Two types:

- Edge couplers: Light enters horizontally from the chip edge. Low loss, wavelength-tolerant. Requires precise alignment — a mechanical precision constraint for packaging.
- Grating couplers: Light couples vertically through a diffraction grating etched into the silicon surface. More compatible with wafer-level testing. Higher loss than edge couplers. Strong wavelength and polarization dependence.

> **PACKAGING IMPACT:** Coupler choice directly affects the packaging strategy and test methodology. TSMC's COUPE platform uses advanced coupler designs as a key architectural element. An architect evaluating a CPO solution must understand which coupler type is used and what alignment tolerance it requires.

Lasers

Silicon does not emit light efficiently due to its indirect bandgap. Laser sources in silicon photonics platforms are either: (a) external lasers coupled into the PIC via fiber or waveguide, or (b) heterogeneously integrated III-V lasers (InP, GaAs) bonded onto the silicon substrate. The laser is the highest-reliability-risk component in an optical link. From an architect's view, laser sourcing strategy (on-package vs. off-package) affects thermal design, power delivery, and redundancy architecture.

> **ARM NEOVERSE:** ARM-based datacenter SoCs have historically had no optical interface. As CPO integrates into Neoverse platform packages, the question of laser placement (co-packaged vs. external) becomes a first-order architectural decision affecting thermal, mechanical, and power delivery design — all directly in the architect's scope.

Chapter 3: Photonic Integrated Circuits (PICs) — System Perspective

3.1 What is a PIC

A Photonic Integrated Circuit (PIC) integrates multiple optical components on a single chip: waveguides, modulators, photodetectors, couplers, and supporting electronics. The PIC is to optics what an ASIC is to logic — it replaces discrete optical components with a fabricated monolithic device.

Silicon PICs are fabricated on Silicon-on-Insulator (SOI) wafers using CMOS-compatible processes. This compatibility with existing semiconductor manufacturing is why silicon photonics is cost-competitive at scale.

3.2 PIC Architecture Concepts the System Architect Uses

Wavelength Division Multiplexing (WDM)

WDM transmits multiple data streams simultaneously on a single waveguide, each on a different wavelength (color of light). A 4-wavelength WDM system multiplies per-fiber bandwidth by 4x without increasing fiber count. From a system architect's view, WDM determines how bandwidth scales across a link: more wavelengths, more bandwidth density, but also more thermal management complexity (each wavelength must be individually controlled).

Optical Link Budget

The optical link budget is the balance of optical gain and loss across a transmission path. Every component — coupler, waveguide, splitter, connector — contributes loss (in dB). The laser provides gain (optical power). The link works if total loss < total gain + receiver sensitivity margin. Architects use link budgets the way they use power budgets: as a constraint that gates design decisions.

Link Element	Typical Loss (dB)	Architect Action if Exceeded
Grating coupler (in/out)	2–5 dB each	Switch to edge coupler
Waveguide routing (per cm)	2–3 dB/cm	Reduce PIC footprint or use lower-loss SiN platform
Fiber connector	0.5–1 dB	Improve alignment tolerance spec
Optical splitter (1x2)	3–4 dB	Minimize splits; increase laser power

Electronic-Photonic Co-Design

Modern silicon photonics platforms integrate driver electronics on the same chip or in the same package as the PIC. The driver ASIC controls the modulator voltage, bias currents, and, critically, thermal tuning of ring resonators. This co-design means the architect must treat the

optical engine as an electronic + photonic system — with shared power delivery, shared thermal constraints, and shared firmware/software control surfaces.

> **INTERFACE BOUNDARY:** The boundary between the driver ASIC and the PIC is an architectural interface. Specification of that interface — power delivery, control signals, thermal limits, reset behavior — is firmly in the system architect's scope.

Chapter 4: Co-Packaged Optics (CPO) — The Architecture Shift

4.1 What CPO Is

Co-Packaged Optics (CPO) integrates optical transceivers directly with switch ASICs or compute dies inside the same IC package. The optical engine — containing the PIC, driver electronics, and often the laser — is a chiplet or module co-located with the host ASIC on the same substrate.

This contrasts with the traditional model, where optical transceivers are pluggable modules that connect to the board via an electrical interface (QSFP-DD, OSFP, etc.). In that model, the electrical path from the switch ASIC to the transceiver module can span 14–16 inches of PCB trace — a signal integrity challenge at 112G and 224G SerDes speeds that mandates DSPs.

4.2 CPO vs. Pluggable: The Architectural Tradeoffs

Dimension	Pluggable (OSFP/QSFP-DD)	Co-Packaged Optics (CPO)
Electrical path to optics	14–16 inches (PCB)	<0.5 inch (in-package)
DSP required	Yes (112G+)	Eliminated
Power efficiency	~5–8 pJ/bit	~1–2 pJ/bit (target)
Field replaceability	Yes (hot-swappable)	No (package-integrated)
Thermal coupling	Isolated (module handles own thermal)	Tightly coupled to ASIC thermal domain
Deployment readiness	Production (current)	Volume production 2026–2028

4.3 TSMC COUPE: The Reference Platform

TSMC's COUPE (Compact Universal Photonic Engine) is the most important reference architecture for understanding CPO integration. It defines how photonic and electronic components co-exist in advanced packaging.

COUPE Generation 1 stacks a 65nm Electronic Integrated Circuit (EIC) with a Photonic Integrated Circuit (PIC) using TSMC's SoIC-X packaging technology. This generation targets OSFP connector applications and delivers 1.6 Tb/s per optical engine.

COUPE Generation 2 moves into CoWoS (Chip-on-Wafer-on-Substrate) packaging, enabling 6.4 Tb/s at the motherboard level. COUPE Generation 3 targets 12.8 Tb/s within processor packages with further reductions in power and latency.

NVIDIA's Quantum-X (InfiniBand) and Spectrum-X (Ethernet) photonic switch platforms follow the COUPE roadmap directly, making COUPE the de facto architectural reference for AI datacenter optical interconnect.

4.4 Architectural Implications of CPO for Platform Architects

Thermal Co-Design

In a CPO package, the PIC and switch ASIC share a thermal domain. Ring resonator modulators require active thermal tuning — heaters consume power to stabilize wavelength. The thermal budget for the optical engine must be co-planned with the ASIC thermal budget. Hot spots from the ASIC can detune ring resonators. This requires thermal simulation at the package level that includes both electronic and photonic die.

Power Delivery

The optical engine (EIC + PIC) adds power delivery requirements to the package power delivery network. Laser bias currents require clean, low-noise power rails. Modulator drivers have specific voltage and current profiles. The architect must define a power delivery architecture that accounts for optical engine requirements — including any isolation requirements between high-speed driver rails and sensitive analog circuits.

I/O Boundary Redefinition

In a traditional SoC, the I/O boundary is electrical: SerDes lanes at a defined voltage and frequency. In a CPO-integrated platform, the I/O boundary becomes optical: a PIC with defined optical interface specifications (wavelength, power, modulation format). The architect must specify and manage both boundaries — the electrical boundary between the host ASIC and the optical engine, and the optical boundary between the optical engine and the fiber plant.

Firmware and Software Surface

CPO adds a new firmware-visible hardware domain. The optical engine requires: laser enable/disable and bias current control, modulator bias and equalization, thermal tuning loop control for ring resonators, link status monitoring (optical power, BER, alarms), and forward error correction (FEC) configuration. These functions are typically exposed through I2C, MDIO, or custom management interfaces. Defining the firmware surface area for optical engine management is an architect-level task.

> **ARM NEOVERSE:** On ARM Neoverse platforms, optical engine management would sit in platform firmware (e.g., SCP or an MCTP-capable management controller), with interfaces exposed through ACPI or device-tree entries to the OS/hypervisor. The SMCCC and SCMI frameworks may need extension to cover optical power management states. This is an active architecture gap as of 2026.

Chapter 5: Industry Architecture Landscape

5.1 Major Players and Their Architectural Approaches

Company	Product/Platform	Architecture	Status (2026)
NVIDIA	Quantum-X / Spectrum-X	CPO with TSMC COUPE; 1.6T–3.2T per switch	Quantum-X 2H 2025; Spectrum-X 2H 2026
Broadcom	Tomahawk 5-Bailly	2nd gen CPO; 100G/lane volume production	Volume production (1st commercial CPO)
Marvell	Custom XPU + CPO	3D SiPho engines on XPU substrate with HBM + chiplets	Announced Jan 2025
Ayar Labs	TeraPHY / UCIe chiplet	Optical chiplet via UCIe for AI scale-up; 8 Tbps	Sampling; GUC integration announced
Lightmatter	Passage M1000 / L200	Optical interconnect for XPU and switch; CPO elements	Announced 2025
STMicro / Intel / IMEC	Silicon photonics platforms	Foundry/process platform suppliers; not system-level products	Active fab capacity

5.2 The UCIe Optical Chiplet Model

Ayar Labs' TeraPHY is the industry's first UCIe (Universal Chiplet Interconnect Express) optical chiplet. UCIe defines the die-to-die interface standard that allows chiplets from different vendors to be integrated on the same substrate. Extending UCIe to cover optical chiplets means optical I/O can be treated architecturally the same as any other die-to-die interface — with defined electrical interface, power envelopes, and mechanical footprints.

For architects, the UCIe optical chiplet model is significant: it means optical I/O bandwidth can be planned as a chiplet resource, allocated and scaled in the same framework used for HBM, compute dies, and I/O dies in a heterogeneous compute package.

> **ARCHITECTURAL SIGNAL:** The UCIe optical chiplet model is the clearest current path toward treating optical interconnect as a first-class architectural element in multi-die AI accelerator packages.

5.3 Roadmap Cadence

Understanding the roadmap cadence is essential for architects who must make platform decisions 3–4 years in advance:

Timeframe	Data Rate	Key Technology	Architecture Implication
2025–2026	100G/lane; 1.6T per engine	COUPE Gen 1; OSFP CPO; Broadcom TH5	CPO at switch layer; compute still electrical
2026–2027	200G/lane; 3.2T per engine	COUPE Gen 2; CoWoS CPO; NVIDIA Spectrum-X	CPO enters motherboard-level integration

2028–2030	400G+/lane; 6.4–12.8T	COUPE Gen 3; in-package optical; UCIe optical chiplets	Optical I/O as first-class die resource; deep package integration

Chapter 6: Platform Architect Decision Framework

6.1 The Four Architecture Decisions

When a system architect encounters a silicon photonics component in a platform design, four decisions govern the architecture:

Decision 1: Integration Point

Where does the optical engine connect to the compute or switch die? Options range from board-level (pluggable module), to package-level (CPO on same substrate), to die-level (3D stacked PIC). Each integration point has different implications for thermal, power delivery, and I/O interface specification.

Decision 2: Laser Strategy

Is the laser co-packaged, on a separate module, or external? Co-packaged lasers simplify the optical path but add thermal load and reliability risk to the package. External lasers are more reliable and field-replaceable but require fiber routing into the package, which introduces mechanical design constraints.

Decision 3: Thermal Architecture

The optical engine requires thermal management. For ring resonator-based engines, active heater power (for wavelength tuning) must be budgeted. For any optical engine, the thermal interface to the package lid or cold plate must be designed. This includes: identifying the TDP contribution of the optical engine, defining its thermal interface material (TIM) and heatsink, and ensuring ASIC thermal gradients do not violate optical engine junction temperature limits.

Decision 4: Management Interface

What firmware or software controls the optical engine? This includes: the physical management interface (I2C, MDIO, PCIe), the control protocol, the power state machine, alarm handling, and exposure to OS/hypervisor through ACPI or device-tree. Defining this interface is an architect-level task that must be done before silicon tape-out.

6.2 Architect Competency Checklist

A system architect working on CPO-integrated platforms should be able to answer the following without reference material:

- What is the optical power budget for this link? Is the margin adequate?
- What is the thermal contribution of the optical engine to the package TDP?
- What are the thermal sensitivity limits for the modulator technology used (ring vs. MZM)?
- What is the management interface to the optical engine and who owns the firmware?
- What is the failure mode if the optical engine loses laser power? How does the system fail gracefully?
- How is the optical engine included in platform power management states (S0, S0ix, etc.)?
- What are the optical engine's power-on and initialization sequences?

> **ARM NEOVERSE:** For ARM Neoverse: which firmware layer (SCP, AP firmware, UEFI/EDK2) owns optical engine initialization? Is there a Trusted Firmware-A (TF-A) handoff? How do SCMI power management messages extend to cover optical domains?

6.3 Failure Mode Analysis — Optical vs. Electrical

System architects familiar with electrical SerDes debug must understand that optical link failures present differently:

Failure Mode	Electrical SerDes	Optical / CPO
Link down	Check voltage levels, termination, CDR lock	Check laser enable, optical power, fiber alignment, receiver sensitivity
High BER	SI analysis: eye diagram, jitter, crosstalk	Optical: OSNR, modulator bias drift, thermal detuning of rings
Intermittent errors	Power supply noise, thermal cycling of PCB	Thermal transient detuning of ring resonators; laser RIN
Power anomaly	Check VR output, decap, load steps	Check laser bias current, heater power; cross-reference thermal readings

Chapter 7: Vocabulary Reference for System Architects

This chapter provides a compact reference for silicon photonics terminology used in architecture documents, vendor datasheets, and standards bodies. Terms are organized by conceptual group.

7.1 Optical Physics Terms

Term	Definition (Architect View)
Wavelength (nm)	Color of light. Silicon photonics uses 1310nm or 1550nm (near-infrared). Determines waveguide design and laser choice.
Refractive index	Material property that determines how fast light travels. The contrast between Si (3.5) and SiO_2 (1.5) creates waveguide confinement.
dB (optical loss)	Logarithmic measure of optical power loss. 3 dB = half the optical power. Used in link budgets. Every component contributes loss.
OSNR	Optical Signal-to-Noise Ratio. The optical analog of SNR. Low OSNR degrades BER. Determined by laser power, amplifier noise, and loss.
Extinction ratio (ER)	Ratio of optical power for logical '1' vs '0'. Higher ER = better signal quality. Modulator design determines ER.

7.2 Device Terms

Term	Definition (Architect View)
PIC	Photonic Integrated Circuit. The optical analog of an ASIC. Contains waveguides, modulators, detectors, and couplers on a single chip.
EIC	Electronic Integrated Circuit. The driver/receiver ASIC co-located with the PIC in an optical engine. Controls modulator bias, laser, thermal tuning.
MZM	Mach-Zehnder Modulator. Large-area, temperature-stable modulator. Used in coherent and long-reach links.
Ring modulator	Compact, low-power modulator. Thermally sensitive. Used in CPO short-reach applications. Requires active thermal tuning.
Ge-on-Si PD	Germanium-on-Silicon photodetector. Standard receiver element in silicon photonics PICs.
SOI	Silicon-on-Insulator. The wafer platform for silicon photonics. A thin silicon layer over a buried oxide (BOX) layer on a bulk silicon substrate.

7.3 System and Packaging Terms

Term	Definition (Architect View)

CPO	Co-Packaged Optics. Optical transceiver integrated in the same IC package as the switch ASIC or compute die. Eliminates board-level electrical interconnect to optics.
COUPE	TSMC's Compact Universal Photonic Engine. Reference CPO platform stacking EIC and PIC via SoIC-X and CoWoS packaging. Three generations targeting 1.6T, 6.4T, 12.8T.
WDM	Wavelength Division Multiplexing. Multiple data streams on different wavelengths over a single fiber/waveguide. Multiplies bandwidth without adding fiber.
CWDM / DWDM	Coarse WDM (wider channel spacing, lower cost) vs. Dense WDM (narrower spacing, higher channel count, requires tighter temperature control).
Optical engine	The combined EIC + PIC unit that performs electrical-to-optical and optical-to-electrical conversion. The functional unit that CPO integrates into the package.
TeraPHY	Ayar Labs' optical I/O chiplet. First UCIe-based optical chiplet for AI scale-up. 8 Tbps bandwidth.
DSP (optical context)	Digital Signal Processor in a pluggable transceiver. Corrects signal degradation over the PCB electrical path. Eliminated by CPO.
FEC	Forward Error Correction. Applied to optical links to recover from bit errors. Adds latency but enables lower-power optical operation. Must be accounted for in latency budgets.

Chapter 8: Learning Path and Study Plan

8.1 Recommended Sequence

The following 8–12 week plan assumes 5–8 hours per week of focused study. It is structured for an engineer with x86 SoC/BIOS background and no prior photonics training.

Week	Topic	Primary Resource	Output
1–2	Optical physics primer + waveguide basics	Chrostowski & Hochberg, Silicon Photonics Design (Ch. 1–3)	Can define loss budget for a simple PIC
3–4	Modulators, detectors, couplers	Chrostowski Ch. 4–6; Lumerical app notes (free)	Can compare MZM vs. ring for a given thermal spec
5–6	CPO architecture + COUPE deep dive	TSMC COUPE papers (IEEE ECTC 2025); NVIDIA Quantum-X tech blog	Can describe COUPE Gen 1–3 architectural tradeoffs
7–8	Industry ecosystem + roadmap	Yole Group Silicon Photonics 2025 report; EDN CPO 2026 article	Can position vendors by architectural approach
9–10	Thermal + power co-design	Ayar Labs CPO integration white papers; Broadcom TH5 packaging notes	Can define thermal budget for an optical engine in a package
11–12	Firmware/management surface	OIF CFP-MSA specs; CMIS standard (Common Management Interface Spec)	Can define firmware interface for an optical engine

8.2 Key Courses

- UBC / edX: Silicon Photonics Design, Fabrication and Data Analysis — Lukas Chrostowski (offered multiple times/year via AIM Photonics). Best introductory structured course available.
- Colorado State ECE544: Silicon Photonics Design — Graduate course using Synopsys and Ansys Lumerical tools. Public syllabus and reading list available online. Good for device-level awareness without full enrollment.
- AIM Photonics Academy (aimphotonics.com): Modular courses on specific topics (photodetectors, modulators, passive devices, fabrication). Self-paced, free for most modules.
- SPIE SC817 short course: Silicon Photonics. Targeted at industry practitioners. Dense, compressed format.

8.3 Key Reading — Primary Sources

- Chrostowski & Hochberg, Silicon Photonics Design, Cambridge University Press, 2015. The field's standard reference. Chapters 1–6 cover everything needed for Role A.
- TSMC COUPE papers, IEEE ECTC 2025. Direct architectural specification of the production CPO platform.
- NVIDIA Technical Blog: A New Era in Data Center Networking with NVIDIA Silicon Photonics-Based Network Switching, April 2025.
- EDN: Where Co-Packaged Optics Technology Stands in 2026, December 2025. Comprehensive industry status review.
- Yole Group: Silicon Photonics 2025 — Focus on SOI, SiN, LNOI & InP Platforms. Market and technology roadmap.
- IEEE JSTQE: Challenges in the Path Toward a Scalable Silicon Photonics Implementation of Deep Neural Networks. Research context for photonic AI acceleration.
- ScienceDirect: The Emerging Applications of Silicon Photonics, December 2025. Survey of applications beyond datacenter: LiDAR, quantum, neuromorphic.
- OIF: Common Management Interface Specification (CMIS). The firmware interface standard for optical modules. Role A architects must be fluent in this document.

8.4 Simulation Tools (Awareness Level, Not Proficiency Required)

Role A architects should be aware of these tools and be able to read output from them, without needing to run simulations independently:

- Ansys Lumerical FDTD / MODE / INTERCONNECT: Industry-standard photonic simulation suite. FDTD for waveguide/coupler physics; MODE for waveguide modes; INTERCONNECT for circuit-level optical simulation.
- Synopsys OptoDesigner / PhoeniX: Used in academic courses (CSU ECE544). PIC layout and simulation.
- Klayout: Open-source IC layout editor. Used for PIC mask layout verification. Free, available on all platforms.

Chapter 9: Standards and Specifications Reference

9.1 Key Standards Bodies

- OIF (Optical Internetworking Forum): Defines electrical and optical interface standards for optical modules and CPO. Key documents: CMIS (Common Management Interface Spec), co-packaged optics white papers, CEI (Common Electrical Interface) specifications.
- IEEE 802.3: Ethernet physical layer standards. Relevant for 400G, 800G, 1.6T Ethernet over optical. Defines PMD (Physical Medium Dependent) sublayer specifications.
- UCIe Consortium: Die-to-die interface standard. Extending to optical chiplets through Ayar Labs' TeraPHY. Defines electrical interface that the optical chiplet presents to the host die.
- Multi-Source Agreements (MSAs): OSFP-XD MSA, QSFP-DD MSA. Define form factors and management interfaces for pluggable transceivers that coexist with CPO during the transition period.

9.2 CMIS — The Firmware Architect's Standard

CMIS (Common Management Interface Specification) defines the management interface protocol for optical modules and optical engines. It specifies the register map, state machine, alarm/warning thresholds, and control sequences that firmware uses to manage optical components.

Key CMIS concepts for architects:

- Module state machine: Defines power-on sequence, low-power mode, data path initialization, and fault handling. Firmware must implement this state machine for each optical engine.
- Data path configuration: Configuration registers for host-to-module electrical interface — lane assignment, FEC mode, application codes.
- VDM (Versatile Diagnostic Monitoring): Optical power readings, temperature, voltage, bias current. The optical engine's equivalent of IPMI sensor data.
- Firmware update: CMIS defines the firmware update procedure for module microcontrollers. Architects must consider how this interacts with platform firmware update workflows (e.g., UEFI capsule update).

9.3 CEI — Electrical Interface Specifications

CEI (Common Electrical Interface) specifications from OIF define the electrical signaling between the host ASIC and the optical engine. For CPO, the relevant specifications are at very short reach (VSR) and ultra-short reach (XSR): CEI-224G-XSR targets <5mm trace length and <1pJ/bit — the electrical interface within a CPO package.

These specifications define the signal integrity requirements at the ASIC-to-optical-engine electrical boundary. Architects use these specs to define die placement and routing rules in the package substrate.

> **ARM NEOVERSE:** ARM Neoverse platforms will require CEI-compatible SerDes in the compute SoC to interface with optical engines. Verifying this compatibility — and negotiating with ARM or the SoC vendor if gaps exist — is an architect-level task for platform bring-up.

Chapter 10: Self-Assessment and Milestone Markers

10.1 Competency Milestones

Use these milestones to self-assess progress. Each milestone maps to a real architecture task.

Level	Can Do	Test Question
L1	Explain CPO vs. pluggable tradeoffs to a non-specialist	Why does NVIDIA's co-packaged photonics claim 3.5x lower power than pluggable transceivers?
L2	Read and critique a CPO vendor datasheet	A vendor claims 1.6T CPO at 2pJ/bit. What questions do you ask to validate the spec?
L3	Define the thermal budget for a CPO module in a package	An optical engine consumes 15W and contains ring modulators with a 0.1nm/K thermal sensitivity. The ASIC junction temperature varies by 20°C under load. What is the thermal risk and how do you mitigate it?
L4	Define the firmware management interface for an optical engine	Sketch the CMIS state machine transitions for optical engine initialization on power-on. What firmware component owns this on an ARM Neoverse platform?
L5	Evaluate CPO solutions against platform requirements	You are architecting a 400G/port switch ASIC platform targeting 2028 production. Compare COUPE Gen 2 and TeraPHY optical chiplet on thermal, power, packaging complexity, and firmware surface. Which do you recommend?

10.2 Knowledge Gaps This Tutorial Does Not Cover

Be explicit about what you will not know after completing this tutorial:

- PIC device design: waveguide mode calculations, FDTD simulation, layout design rules. This is Role B.
- Coherent optics: DSP algorithms, IQ modulation, dispersion compensation. Relevant for long-haul, less so for CPO short-reach.
- Laser physics: rate equations, noise analysis, reliability modeling. Awareness-level covered; design-level is not.
- Optical test and measurement: use of optical spectrum analyzers, BER testers, OTDR. Practical lab skills not covered here.
- Silicon nitride (SiN) and lithium niobate ($LiNbO_3$) platforms: alternative PIC platforms with different loss and bandwidth characteristics. Mentioned but not detailed.

Appendix A: Annotated References

Books

- Chrostowski, L. & Hochberg, M. Silicon Photonics Design. Cambridge University Press, 2015. ISBN 978-1107085459. The primary textbook for Role A study. Chapters 1–6 are essential. Practical focus on design over theory.
- Reed, G.T. (ed.) Silicon Photonics: The State of the Art. Wiley, 2008. Older but foundational. Good for understanding the physics of silicon as an optical medium.
- Farsaei, A. Introduction to Layout Design and Automation of Photonic Integrated Circuits. Springer, 2023. Relevant if you need to read PIC layout documentation.

Industry Papers and Reports

- TSMC COUPE papers, IEEE ECTC 2025. Direct specification of production CPO architecture. Led by VP K.C. Hsu. Covers COUPE, EPIC-BOE, and iOIS platforms.
- NVIDIA Technical Blog: A New Era in Data Center Networking with NVIDIA Silicon Photonics-Based Network Switching. developer.nvidia.com/blog. April 2025. Specification-level description of Quantum-X and Spectrum-X architectures.
- EDN: Where Co-Packaged Optics Technology Stands in 2026. edn.com. December 2025. Comprehensive 2025 ecosystem status. Covers Broadcom, Marvell, Ayar Labs, GUC integration.
- Yole Group: Silicon Photonics 2025 — Focus on SOI, SiN, LNOI & InP Platforms. yolegroup.com. September 2025. Market forecasts and technology roadmaps.
- IEEE JSTQE: Challenges in the Path Toward a Scalable Silicon Photonics Implementation of Deep Neural Networks. DOI: 10.1109/JSTQE.2019.2930455. Research context for photonic AI.
- ScienceDirect: The Emerging Applications of Silicon Photonics. Science Direct (Elsevier), December 2025. Survey of LiDAR, quantum, and neuromorphic photonics applications.
- IEEE Photonics Society: 2025 IEEE Study Leverages Silicon Photonics for Scalable and Sustainable AI Hardware. ieeephotonics.org. April 2025. HPE Labs photonic AI accelerator using III-V/Si heterogeneous integration.
- Tom's Hardware: NVIDIA Outlines Plans for Silicon Photonics by 2026. August 2025. Accessible summary of NVIDIA's three-generation CPO roadmap.
- STMicro Blog: Light into Data: How Silicon Photonics is Powering the AI Data Center Revolution. blog.st.com. November 2025. Good primer on silicon photonics fundamentals from a manufacturing perspective.

Standards Documents

- OIF CMIS (Common Management Interface Specification), latest version. oiforum.com. Essential for firmware architects. Defines the management interface for all optical modules.
- OIF CEI-224G-XSR. oiforum.com. Electrical interface specification for ultra-short reach CPO applications.

- UCIe Specification, Version 1.1. uciexpress.org. Die-to-die interface standard. Relevant for optical chiplet integration.
- IEEE 802.3df: 200 Gb/s, 400 Gb/s, 800 Gb/s, and 1.6 Tb/s Ethernet. Defines optical PMD specifications for datacenter Ethernet.

Courses and Learning Platforms

- UBC / edX / AIM Photonics: Silicon Photonics Design, Fabrication and Data Analysis. Prof. Lukas Chrostowski. aimphotonics.com/online-courses. Offered multiple times/year. Best available structured course.
- Colorado State ECE544: Silicon Photonics Design. engr.colostate.edu/~mnikdast/silicon-photonics-design/. Public syllabus and reading list. Uses Synopsys and Ansys Lumerical.
- AIM Photonics Academy. aimphotonics.com/online-courses. Modular courses: Fundamentals of Integrated Photonics, Photodetectors and Modulators, Passive Devices, Simulation Library. Most modules free.
- SPIE SC817: Silicon Photonics. spie.org/education/courses. Short course format. Industry practitioner focus. Offered at SPIE Photonics West and SPIE Optics + Photonics.
- OFC Short Course SC432: Hands-on Silicon Photonics Components. ofcconference.org. Includes actual PIC fabrication run via UBC. Requires laptop with Lumerical pre-installed.

Key Blogs and Ongoing Sources

- SemiVision Research (Substack: tspasemiconductor.substack.com): Best ongoing source for TSMC photonics architecture, COUPE updates, and NVIDIA CPO roadmap. Paid tier has deep technical analysis.
- Ayar Labs Blog (ayarlabs.com/blog): Primary source for TeraPHY and UCIe optical chiplet developments.
- IEEE Photonics Society News (ieeephotonics.org): Research-level developments and conference summaries.
- PhotonDelta (photondelta.com): European silicon photonics ecosystem news. Good for foundry and supply chain developments.

Appendix B: Acronym Registry

Acronym	Full Form and Context
APD	Avalanche Photodiode. High-sensitivity photodetector used in some silicon photonics receivers.
BER	Bit Error Rate. Fraction of received bits in error. Key optical link quality metric.
BOX	Buried Oxide. The SiO_2 layer under the active silicon in SOI wafers. Provides optical

	confinement.
CDR	Clock and Data Recovery. Recovers clock from a serial data stream. Used in electrical receivers.
CEI	Common Electrical Interface. OIF standard for host-to-module electrical signaling.
CMIS	Common Management Interface Specification. OIF standard for optical module management firmware interface.
CMN	Coherent Mesh Network. ARM's interconnect fabric for Neoverse SoCs (CMN-700, CMN-700AE).
CPO	Co-Packaged Optics. Integration of optical transceiver within the IC package of the host ASIC.
COUPE	Compact Universal Photonic Engine. TSMC's CPO platform (EIC + PIC via SoIC-X/CoWoS).
CoWoS	Chip-on-Wafer-on-Substrate. TSMC advanced 2.5D packaging technology.
CWDM	Coarse Wavelength Division Multiplexing. WDM with wider channel spacing; lower cost.
DSP	Digital Signal Processor. In optical context: the signal-recovery chip in pluggable transceivers. Eliminated by CPO.
DWDM	Dense Wavelength Division Multiplexing. High channel count WDM; requires tighter thermal control.
EIC	Electronic Integrated Circuit. The driver/receiver ASIC co-located with the PIC in an optical engine.
ER	Extinction Ratio. Optical power ratio between logical '1' and '0'. Higher is better.
FEC	Forward Error Correction. Error correction applied to optical links; adds latency.
FDTD	Finite-Difference Time-Domain. Electromagnetic simulation method used in photonic device design.
GUC	Global Unichip Corp. TSMC-affiliated ASIC design services company; Ayar Labs integration partner.
HBM	High Bandwidth Memory. 3D-stacked DRAM used in AI accelerators. Packaged on same substrate as compute die.
InP	Indium Phosphide. III-V semiconductor used for high-efficiency lasers integrated into silicon photonics PICs.
MZM	Mach-Zehnder Modulator. Temperature-stable, large-footprint optical modulator.
OCS	Optical Circuit Switch. Switches optical signals without electrical conversion.
OIF	Optical Internetworking Forum. Standards body for optical interface specifications.
OSNR	Optical Signal-to-Noise Ratio. Key link quality metric; determined by laser power and amplifier noise.
OSFP	Octal Small Form-Factor Pluggable. Pluggable optical transceiver form factor for 400G/800G.
PDN	Power Delivery Network. The power distribution architecture from VR to die.

PIC	Photonic Integrated Circuit. Monolithic chip integrating optical components.
PSCI	Power State Coordination Interface. ARM firmware standard for CPU power management.
QSFP-DD	Quad Small Form-Factor Pluggable Double Density. High-density pluggable transceiver form factor.
RIN	Relative Intensity Noise. Laser noise metric. High RIN degrades optical signal quality.
SCMI	System Control and Management Interface. ARM firmware standard for platform resource management.
SMCCC	SMC Calling Convention. ARM standard for secure monitor calls in TrustZone.
SoIC-X	TSMC System on Integrated Chips (face-to-face bonding variant). Used in COUPE Gen 1.
SOI	Silicon-on-Insulator. Wafer platform for silicon photonics PICs.
TF-A	Trusted Firmware-A. ARM's reference open-source secure firmware for ARMv8-A/v9-A platforms.
TIM	Thermal Interface Material. Material between a die and its heatsink/cold plate. Determines thermal resistance.
TDP	Thermal Design Power. Maximum sustained power a cooling solution must handle.
UCIe	Universal Chiplet Interconnect Express. Die-to-die interface standard enabling multi-vendor chiplet integration.
VDM	Versatile Diagnostic Monitoring. CMIS framework for optical module sensor data readout.
VR	Voltage Regulator. Power conversion component; in CPO context, must supply clean rails for laser and modulator drivers.
WDM	Wavelength Division Multiplexing. Transmitting multiple data streams on different wavelengths over a single fiber.
XSR	Extra-Short Reach. CEI designation for ultra-short electrical interfaces within CPO packages (<5mm).

Chapter 11: LinkedIn Post Series — Silicon Photonics for Architects

This chapter contains five LinkedIn posts designed for an engineer audience. The posts follow the same arc as this tutorial: from the problem statement through to CPO architecture and ARM platform implications. Each post is self-contained and numbered for sequential publication. Writing style: researcher-oriented, discussion tone, specific language, numbered citations, no hyphens, no dramatic language.

> **ARCHITECT NOTE:** Each post is calibrated for a VP-level technical hiring audience evaluating an architect who has deliberately built this knowledge domain. The goal is to demonstrate depth and architectural judgment, not to explain basics.

Post 1: The Electrical Wall — Why Copper Loses at Scale

> **Post 1 of 5: Silicon Photonics for Platform Architects**

A back-of-the-envelope that changed how I think about AI datacenter architecture.

Scale a GPU cluster to one million nodes. Each GPU needs six pluggable optical transceivers. Each transceiver contains a DSP chip that consumes roughly 30W. That is 180MW of power spent on signal cleanup alone, before a single floating point operation runs.

The DSP exists because copper traces degrade signals. At 112G SerDes speeds, a board-level trace of 14 to 16 inches introduces enough loss and jitter that the receiver cannot decode the data without correction. At 224G, the problem is worse. The physics does not improve with faster process nodes.

Silicon photonics eliminates the DSP by moving the optical interface inside the IC package. The electrical path from switch ASIC to optical transceiver drops from 16 inches to under half an inch. The signal does not degrade. The DSP is not needed.

NVIDIA calls this co-packaged optics. Their Quantum-X and Spectrum-X platforms, targeting 2025 to 2026, claim 3.5x lower power per transceiver compared to pluggable modules. [1] Broadcom's Tomahawk 5-Bailly is the first volume-production CPO solution. [2] TSMC's COUPE platform defines the packaging architecture that makes this possible at manufacturing scale. [3]

For architects, the implication is not just lower power. It is a structural change to where the I/O boundary sits, what the thermal map looks like, and which firmware components own the optical management domain.

That is the thread I am working through this week.

[1] NVIDIA Technical Blog, April 2025. developer.nvidia.com/blog [2] EDN, December 2025. edn.com/where-co-packaged-optics-cpo-technology-stands-in-2026 [3] SemiVision Research / TSMC ECTC 2025. tspasemiconductor.substack.com

Post 2: Ring Resonators and Thermal Budgets

Post 2 of 5: Silicon Photonics for Platform Architects

The modulator is where the data gets onto the light. The choice of modulator technology is also, unexpectedly, a thermal architecture decision.

Two dominant modulator types in silicon photonics.

Mach-Zehnder Modulators (MZMs) are large, around 1mm footprint, and temperature-stable. They encode data by splitting light, phase-shifting one arm, and recombining. Used in coherent systems and long-reach links. Thermal sensitivity is low. An architect can treat an MZM-based link much like an electrical SerDes lane for thermal planning purposes.

Ring resonator modulators are compact, around 10 micrometers in diameter, and power-efficient. Co-packaged optics solutions for short-reach datacenter links prefer rings because of their size and energy advantages. The problem: a 1 degree Celsius temperature change detunes a ring resonator enough to cause a link error.

This is not a device-level problem. It is a platform-level problem.

The switch ASIC sitting next to the optical engine in a CPO package runs hot and varies its power dissipation with workload. The thermal gradient that results is not static. The optical engine's ring resonators need active heaters to compensate. Those heaters consume power, which is part of the optical engine TDP. That TDP must be co-planned with the ASIC thermal budget from the start of the packaging architecture.

An architect who inherits a thermal spec from the ASIC team and then adds an optical engine on top will find the thermal map is wrong. The optical engine thermal domain must be part of the initial package-level thermal simulation, not a post-integration correction.

This is the kind of constraint that looks like a photonics problem but is actually an architecture coordination problem.

[1] Chrostowski & Hochberg, Silicon Photonics Design, Cambridge UP, 2015. Chapter 5. [2] Colorado State ECE544 course notes, engr.colostate.edu/~mnikdast/silicon-photonics-design

Post 3: CPO Is a Packaging Architecture Decision

Post 3 of 5: Silicon Photonics for Platform Architects

Co-packaged optics gets framed as an interconnect technology. It is more precisely a packaging architecture decision, with thermal, power delivery, and I/O boundary consequences that the system architect owns.

TSMC's COUPE platform defines the reference architecture. Generation 1 stacks an Electronic Integrated Circuit (EIC) with a Photonic Integrated Circuit (PIC) using SoIC-X bonding. This is the optical engine. It delivers 1.6 Tb/s per engine and connects to OSFP connectors. Generation 2 moves the optical engine into CoWoS packaging alongside the switch ASIC, at 6.4 Tb/s. Generation 3 targets 12.8 Tb/s within the processor package itself. [1]

Each generation represents a different integration depth. The architectural decisions change with each step.

At Generation 1, the optical engine is an external module. The firmware boundary is similar to managing a pluggable transceiver. Thermal domains are partially isolated. At Generation 2, the optical engine is on the same substrate as the ASIC. Thermal domains merge. The electrical interface between ASIC and optical engine must conform to CEI-224G-XSR: under 5mm trace length, under 1 pJ/bit. [2] The power delivery network must supply clean rails to the laser drivers without crosstalk from the ASIC core.

At Generation 3, the optical engine is inside the processor package. There is no longer a clear boundary between the compute die and the optical domain. The architect must define where the optical management firmware lives, how it maps to platform power states, and what happens to the optical link during a CPU low-power transition.

This progression is not incremental refinement. Each generation requires a different architecture model.

[1] TSMC COUPE papers, IEEE ECTC 2025. Analysis via SemiVision Research. tspasemiconductor.substack.com [2] OIF CEI-224G-XSR specification. oiforum.com

Post 4: The Firmware Surface Nobody Has Defined Yet

Post 4 of 5: Silicon Photonics for Platform Architects

The industry has a detailed standard for managing pluggable optical modules. It is called CMIS: the Common Management Interface Specification. It defines the register map, state machine, alarm thresholds, and firmware update procedure for every optical module. [1]

When the optical engine moves inside the processor package, CMIS still applies to the management interface. But the question of which firmware component owns that interface has not been settled for ARM-based datacenter platforms.

On an ARM Neoverse platform today, platform management sits primarily in the System Control Processor (SCP), with SCMI as the communication protocol between SCP and application processors. [2] SCMI defines power, performance, clock, sensor, and reset domains. It does not define optical domains.

The optical engine requires: laser enable sequencing at power-on, modulator bias initialization, thermal tuning loop control for ring resonators, continuous optical power and BER monitoring via VDM (Versatile Diagnostic Monitoring in CMIS), alarm escalation, and firmware update for the EIC microcontroller.

None of these are covered by the current SCMI specification. Trusted Firmware-A (TF-A) has no optical domain primitives. UEFI/EDK2 has no standard optical engine driver class.

This is an open architecture problem. Someone needs to define which firmware layer owns optical engine initialization, how optical power states map to ACPI device power states, and how CMIS state machine transitions integrate with platform boot sequencing.

The answer will likely vary by platform vendor until a standards body formalizes it. The architect who defines this cleanly for their platform has a durable advantage.

[1] OIF CMIS specification. oiforum.com [2] ARM SCMI specification DEN0056E v3.2. developer.arm.com/documentation

Post 5: What an Architect Needs to Know — and What They Do Not

Post 5 of 5: Silicon Photonics for Platform Architects

I have spent the past several weeks building a working understanding of silicon photonics from the platform architect's perspective. This post is a calibration on where the line actually sits.

What an architect must know.

Optical link budgets: the loss across every component in the optical path, in dB, and whether the laser power covers it with margin. This is the optical equivalent of a power delivery IR drop budget. Architects do link budgets for electrical interfaces. They need to do them for optical interfaces too.

Modulator technology selection and its thermal consequences: MZM is temperature-stable; ring resonators require active thermal tuning and impose constraints on the package thermal gradient.

CPO integration depth and its effect on firmware surface: COUPE Gen 1 through 3 require progressively different approaches to thermal co-design, power delivery, and management interface ownership.

CMIS at the interface level: what the state machine requires, what VDM provides, and how firmware update for the optical engine microcontroller fits into platform update workflows.

What an architect does not need to know.

Waveguide mode calculations. FDTD simulation. PIC layout design rules. Laser rate equations. These are device design tasks. They require years of electromagnetics background. Architects who need to understand a CPO vendor's design choices can read the specifications produced by that work without needing to produce it themselves.

The distinction matters because the learning investment is very different. Device design fluency takes years. System-level CPO architecture fluency, for an engineer with SoC and platform background, takes months.

Large-scale CPO deployment is projected between 2028 and 2030. [1] The architects who will define those platforms are being hired now. The technical depth required to do that work sits at the intersection of packaging, thermal, power delivery, firmware, and optical systems. Most people in this industry have depth in one or two of those domains. Very few have a working model across all five.

That is the gap I am working to close.

[1] Yole Group, Silicon Photonics 2025. yolegroup.com

End of Document — Silicon Photonics for the System Architect v1.1